Skill discovery trial

Five OpenRouter models | Four methods | A small, real integration test

Executive Summary

We tested how five models choose useful skills from the pinned SkillRet dataset, using LangChain Deep Agents' real skill middleware. The trial compared listing every skill, keyword search, meaning-based search and category-first search on three requests, with libraries of 50, 250 and 1,000 skills. The integration worked, but only 24 selections matched the reference answers exactly. This sample is too small to choose a winning model.

180
planned cases
138 attempted; 42 skipped

121
valid selections
Correct format and allowed IDs

24
exact matches
All and only the reference skills

Results by model

Each model had 36 planned cases. A valid selection can still be wrong. Exact counts below use only valid selections; coverage differs.

Model	Valid / attempted	Exact / valid	Coverage note
Qwen 3.8 27B	2 / 5	2 / 2	Mostly paused: rate limits
Qwen 3.8 Flash	34 / 35	2 / 34	Nearly complete
DeepSeek V4.1 Flash	34 / 35	8 / 34	Nearly complete
MiMo-V2.6-Pro	17 / 28	4 / 17	Invalid IDs; provider overload
GLM-5.3-Flash	34 / 35	8 / 34	Nearly complete

Of 180 cases, 53 reuse earlier matching runs. The 59 cases without valid selections comprise 13 invalid-ID outputs, 4 provider errors, 32 provider-paused cases and 10 prerequisite-gated cases.

Offline baseline: 51.67% Recall@5

Before paid calls, full-document BM25 found an average 51.67% of the annotated skills in its top five results: 200 queries against 6,006 skills. This is a different test from the metadata-only selections above.

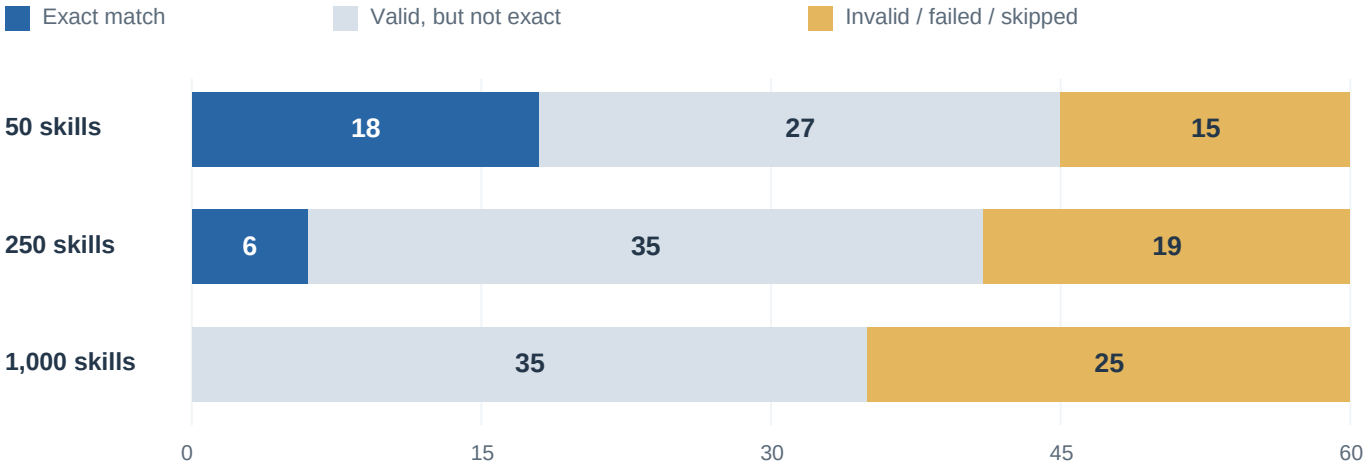
Cost: expansion US$0.3379; whole project US$0.3865 (estimated CAD$0.5797 at 1.50). Costs are provider-reported. The report itself used no paid model calls.

Results in two graphs

Selection quality and text volume answer different questions.

Selection outcomes by library size

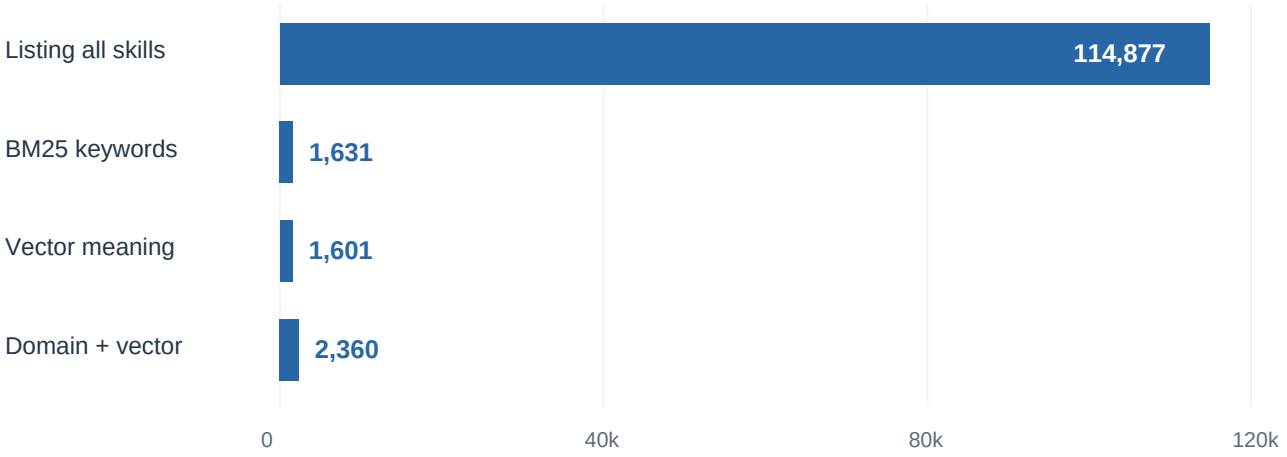
60 planned cases per row: 5 models x 3 requests x 4 methods. Counts, not percentages.



At 1,000 skills, **0 of 35 valid selections were exact**. Still, 22 included a reference skill, while 34 included extra IDs. These three requests do not establish a reliable scaling trend, and failures change which models contribute to each row.

Text processed at 1,000 skills

DeepSeek V4.1 Flash example | Mean input + output tokens per workflow



Tokens are pieces of text processed by a model. In this example, keyword search used about **1/70 of the tokens** needed to list every skill. Lower text volume alone does not prove better selections or an equal reduction in billed cost.

Sample sizes: 3 completed responses each for listing, BM25 and vector; 2 for domain-vector. Includes invalid responses with usage data and category-routing tokens, even when reused. This is workflow text volume, not newly billed traffic.

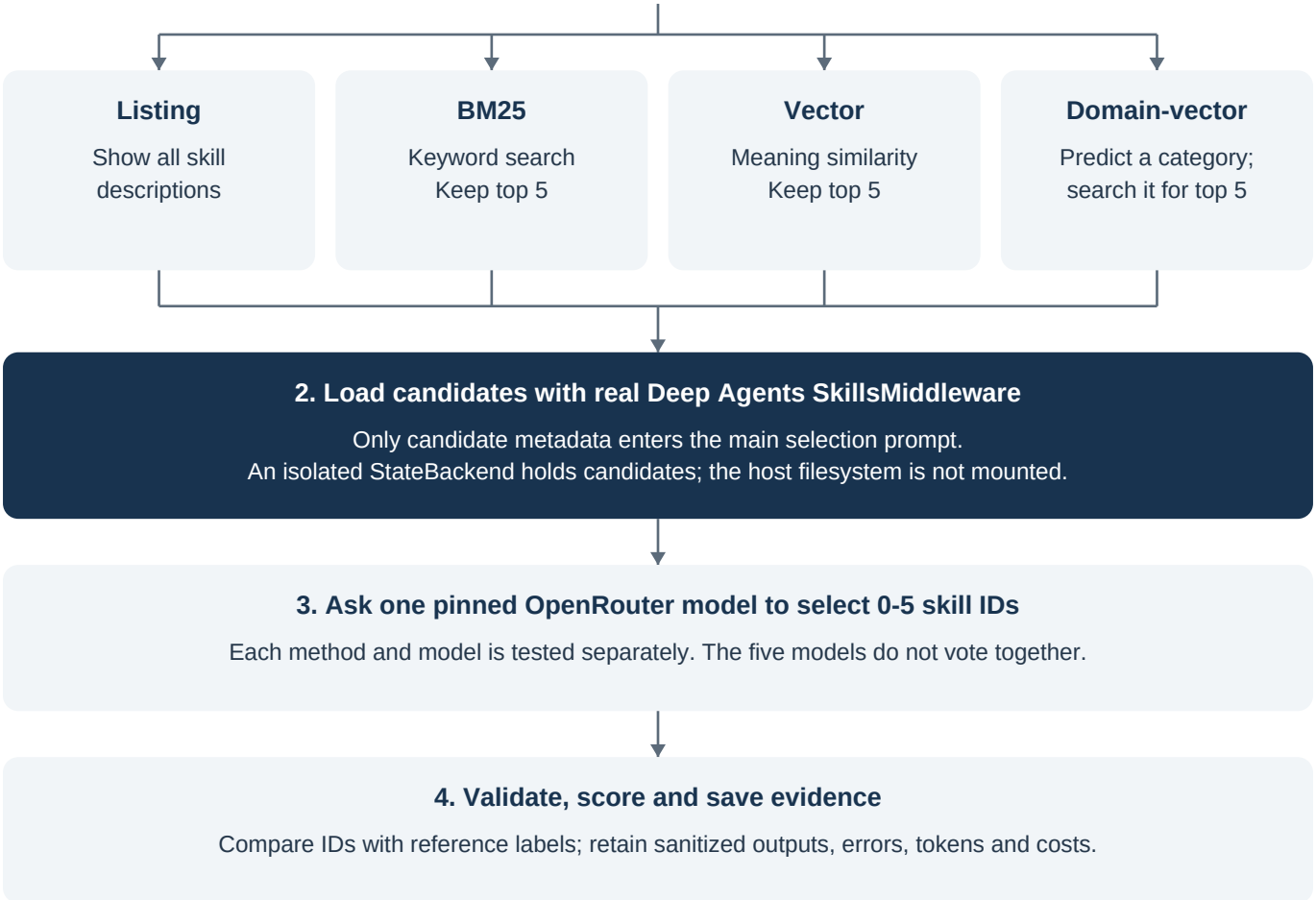
How the test works

The model chooses skill IDs; downloaded skill instructions are never executed.

1. Prepare the same test library and request

Pinned SkillRet data | 50, 250 or 1,000 skills | Three fixed request texts

Reference labels keep relevant skills in each test library and are used for scoring. They never enter model prompts or model state.



Controls: all model calls require local paid-run permission and a persistent budget guard. Caps: US$3 / CAD$5 and 300 requests. Providers are pinned; no silent fallback or context truncation. Category-first cases run only after prerequisite checks pass.

What was verified separately

All five models passed a live integration canary. The earlier activation check recorded **16 actual native SKILL.md reads across 25 cases**; 15 cases completed the workflow. A successful read is not proof that a skill can execute a task.

Next step: resolve provider availability, then test more fixed requests with comparable coverage. The present results support integration and cost observations, not a general model ranking.

Evidence: reports/OPENROUTER_SCALING.json; results/openrouter/scaling/trial.jsonl; results/baseline_summary.jsonl; reports/OPENROUTER_MODELS.json.
SkillRet revision: 6583d7d2ed07644d0fb8938ed8178f3a7dc42a12